17C 1 HETEROGENEOUS CATALYSIS

SPECIFICATION REFERENCE

17.26 PART	17.27	17.28
	17.29(i)	17.29(ii)

LEARNING OBJECTIVES

- Know that transition metals and their compounds can act as heterogeneous catalysts.
- Know that a heterogeneous catalyst is in a different phase from the reactants and that the reaction occurs at the surface of the catalyst.
- Understand, in terms of oxidation number, how V_2O_5 acts as a catalyst in the Contact process.
- Understand how a catalytic converter decreases carbon monoxide and nitrogen monoxide emissions from internal combustion engines by:
  - (i) adsorption of CO and NO molecules onto the surface of the catalyst, resulting in the weakening of bonds and chemical reaction
  - (ii) desorption of CO_2 and N_2 product molecules from the surface of the catalyst.

DIFFERENT WAYS TO UNDERSTAND CATALYSIS

When you first learned about catalysts you probably recognised them as substances that increased the rate of a reaction and remained chemically unchanged at the end of the reaction. This description is fine as an introduction to catalysis, but it does not tell you anything about how catalysts work.

In **Topic 9 (Book 1: IAS)** you learned about catalysis in terms of its effect on the activation energy of a reaction. You also learned how to represent catalytic action using Maxwell–Boltzmann energy distributions and energy profiles.

In this section and the next one, we will consider catalysis in more detail by looking at the two main types of action: heterogeneous and homogeneous.

TRANSITION METALS AS HETEROGENEOUS CATALYSTS

A **heterogeneous catalyst** is one that is in a different phase from that of the reactants. You can probably remember an example of this type of catalysis from many years ago. Oxygen gas can be made in the laboratory by decomposing a hydrogen peroxide solution. The catalyst used in this reaction is usually manganese(IV) oxide – note that this is a compound of a transition metal. However, what makes this an example of heterogeneous catalysis is that manganese(IV) oxide is a solid and hydrogen peroxide solution is a liquid.

Many transition metals, and their compounds, are used as solid catalysts. Their action can be explained in terms of what happens on the surface of the catalyst. The fact that the action takes place only on the surface explains why many of them are used in a finely divided form, as small particles (including powder) rather than as large lumps. Sometimes, they are used instead as a thin coating on an inert support material.

EXAM HINT

Industrial chemists often use heterogeneous catalysts rather than homogeneous catalysts. This is because it is easier to separate the reaction products from the catalyst if they are in different phases.

THE CONTACT PROCESS

Sulfuric acid is one of the most widely used chemicals, both in terms of the quantity produced and the number of industries that use it. Its biggest single use is in the manufacture of fertilisers. A complete description of the Contact process used to manufacture sulfuric acid is beyond the aims of this book, but the key reaction in the process is the conversion of sulfur dioxide to sulfur trioxide in this reaction:

$$2SO_2 + O_2 \rightleftharpoons 2SO_3$$

At the temperatures and pressures used in the process, all of the substances are in the gas phase, and the mixture of reactants is passed over a catalyst of vanadium(V) oxide, V_2O_5, usually known in industry as vanadium pentoxide (**fig A**).



▲ **fig A** Sample of vanadium(V) oxide, as used as the catalyst in the Contact process.

SURFACE ADSORPTION THEORY

This theory is often used to explain the way that a heterogeneous catalyst works. It is usually considered as having three steps:

1 **Adsorption**, in which one or more reactants become attached to the surface of the catalyst.

2 Reaction, following the weakening of bonds in the adsorbed reactants.

3 **Desorption**, in which the reaction product becomes detached from the surface of the catalyst.

In the Contact process, the reaction step has two parts:

Part 1: sulfur dioxide adsorbs onto the vanadium(V) oxide and a redox reaction occurs:

$$V_2O_5 + SO_2 \rightarrow V_2O_4 + SO_3$$

Note that the oxidation number of vanadium decreases from +5 to +4. The sulfur trioxide then desorbs.

Part 2: oxygen reacts with the V_2O_4 on the surface of the catalyst and another redox reaction occurs:

$$V_2O_4 + \tfrac{1}{2}O_2 \rightarrow V_2O_5$$

Note that the original catalyst is regenerated as the oxidation number increases from +4 to +5.

CATALYTIC CONVERTERS

THE PROBLEMS

One of the major problems associated with increased road vehicle usage over several decades is the increased amount of pollution from vehicle exhaust gases. In some cities, the quality of air breathed by humans is below acceptable levels. Without catalytic converters, the situation would now be much worse.

Although many different pollutants come from vehicle exhausts, two of the most significant are carbon monoxide and nitrogen monoxide.

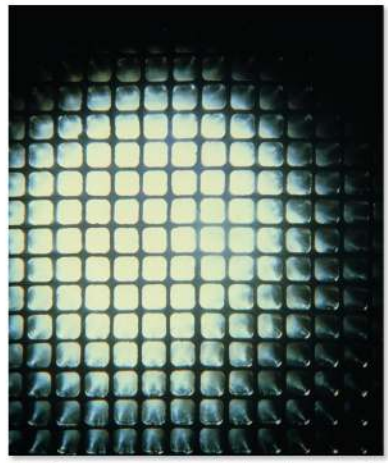
- Carbon monoxide is a toxic gas that interferes with oxygen transport from the lungs through the bloodstream to vital organs in the body (see **Section 17A.6**).
- Nitrogen monoxide is easily oxidised in the atmosphere to nitrogen dioxide. It can act as a respiratory irritant and contribute to the formation of acid rain.

Carbon monoxide forms through the incomplete combustion of hydrocarbon fuels. Nitrogen monoxide forms through the reaction between nitrogen and oxygen at the high temperatures that exist in an internal combustion engine.

THE SOLUTIONS

For over 20 years, cars sold worldwide have been fitted with catalytic converters in an attempt to reduce the effect of vehicle emissions on air quality. The main transition metals used in catalytic converters are platinum and rhodium, and sometimes palladium (**fig B**). The method of action can be explained by the surface adsorption theory described above. In one type of catalyst, molecules of carbon monoxide and nitrogen monoxide are adsorbed onto the surface. Then, because their bonds are weakened, they react together to form carbon dioxide and nitrogen, which are then desorbed from the surface of the catalyst. The overall reaction can be represented by this equation:

$$2CO + 2NO \rightarrow 2CO_2 + N_2$$



▲ **fig B** View through the element of a catalytic converter from a car exhaust. The inner surface is coated with an alloy containing platinum, rhodium and palladium.

LEARNING TIP

Do not confuse absorb/absorption with adsorb/adsorption. Very simply, absorption involves one substance becoming distributed throughout another (for example, water in a sponge), while adsorption only happens at the surface of a substance.

CHECKPOINT

SKILLS CREATIVITY, ETHICS

1. How does the description of the catalytic role of vanadium(V) oxide show that this statement is not true?

   'A catalyst increases the rate of a chemical reaction but does not take part in the reaction.'

2. Write an equation to show how nitrogen monoxide can form acid rain.

SUBJECT VOCABULARY

heterogeneous catalyst a catalyst that is in a different phase from the reactants

adsorption the process that occurs when reactants form weak bonds with a solid catalyst

desorption the process that occurs when products leave the surface of a solid catalyst

17C 2 HOMOGENEOUS CATALYSIS

SPECIFICATION REFERENCE
17.26 PART 17.30 17.31 17.32

LEARNING OBJECTIVES

- Know that transition metals and their compounds can act as homogeneous catalysts.
- Know that a homogeneous catalyst is in the same phase as the reactants and appreciate that the catalysed reaction will proceed via an intermediate species.
- Understand the role of Fe^{2+} ions in catalysing the reaction between I^- and $S_2O_8^{2-}$ ions.
- Know the role of Mn^{2+} ions in autocatalysing the reaction between MnO_4^- and $C_2O_4^{2-}$ ions.

TRANSITION METALS AS HOMOGENEOUS CATALYSTS

A **homogeneous catalyst** is one that is in the same phase as the reactants. This means that they are either all gases, or more often, all in aqueous solution. You have not seen many examples of this type of catalyst, compared with those that take part in heterogeneous catalysis. Homogeneous catalysis is also much less common in industry.

The key feature of homogeneous catalysis is the formation of an intermediate species for which a specific formula can be written.

A REACTION OF THE $S_2O_8^{2-}$ ION

The ion with this formula has several names, the simplest of which is the 'persulfate ion', although it is often known as the 'peroxydisulfate ion'. We will not use the IUPAC name, which is very complicated!

It acts as an oxidising agent in its reaction with iodide ions, the equation for which is:

$$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$$

One reason why this reaction is slow at room temperature is that the two reactant ions are both negatively charged and so repel each other. The reaction is much faster in the presence of Fe^{2+} ions, which act as a catalyst. All of the reactants and products, and the catalyst, are in the aqueous phase, so this reaction is an example of homogeneous catalysis.

STEPS IN THE CATALYSED REACTION

Step 1: The Fe^{2+} ions are not repelled by the $S_2O_8^{2-}$ ions because they have the opposite charge. They react together as follows:

$$S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$$

Step 2: The Fe^{3+} ions formed in Step 1 now react with the I^- ions (these also have opposite charges) as follows:

$$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$$

The iron(II) ions are used in Step 1 and regenerated in Step 2, so the two steps can repeat continuously.

ALTERNATIVE MECHANISM

The reaction is also catalysed by Fe^{3+} ions, for which the following mechanism can be written.

Step 1: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$

Step 2: $S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$

The same two reactions are involved, but they occur in a different order.

OXIDATION OF ETHANEDIOATE IONS

You may know about titrations in which potassium manganate(VII) in acidic conditions acts as an oxidising agent. For these titrations to work well enough, it is important that the reactions are fast, so that the end point of the reaction can be accurately observed (**fig A**).

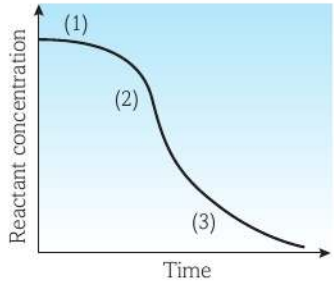
One example of a redox reaction that can be used in titrations is the oxidation of ethanedioate ions by potassium manganate(VII). The equation for this reaction is:

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 5CO_2 + 8H_2O$$

As with the $S_2O_8^{2-}/I^-$ reaction, the reacting species are both negatively charged. Therefore, the reaction is slow. However, as more potassium manganate(VII) solution is added, the reaction rate increases.



▲ **fig A** (a) At the start of the titration, the colour of the potassium manganate(VII) solution takes some time to disappear; (b) the colour of the potassium manganate(VII) solution now disappears more quickly because there are Mn^{2+} ions to catalyse the reaction; (c) at the end of the titration, there are no more ethanedioate ions left to react, so the remaining potassium manganate(VII) solution gives the mixture a pink colour.



▲ **fig B** (1) The slow decrease in the reactant concentration is because the reaction is uncatalysed at first. (2) The reactant concentration decreases more rapidly as the reaction rate increases because of catalysis. (3) The reactant concentration then decreases more slowly because there is little reactant left.

Note that one of the reaction products is the Mn^{2+} ion. This can act as a catalyst in the reaction, which explains why the reaction rate increases as the titration proceeds. The formation of a reaction product that increases the rate of the reaction is sometimes referred to as **autocatalysis**. This effect is sometimes represented using a graph (**fig B**).

LEARNING TIP

Note that in **Section 17C.1**, all of the transition metal catalysts were solids, but that in this section they are all ions in aqueous solution.

CHECKPOINT

1. State why Mg^{2+} ions do not act as a catalyst in the reaction between $S_2O_8^{2-}$ and I^- ions.
2. State why autocatalysis of the reaction between MnO_4^- and $C_2O_4^{2-}$ ions does not occur under alkaline conditions.

SUBJECT VOCABULARY

homogeneous catalyst a catalyst that is in the same phase as the reactants
autocatalysis when a reaction product acts as a catalyst for the reaction